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1. Sebuah benda bergerak dengan percepatan $a(t) = t + 2$ cm/det^2. Diketahui pada saat $t = 1$ detik, kecepatan benda 2,5 cm/det dan posisi benda berada pada jarak 6 cm

Tentukan :

a. Persamaan posisi / jarak benda, s sebagai fungsi waktu t

$a(t) = \frac{dv}{dt}$ $\quad dv = a(t)\,dt$

$v(t) = \int a(t)\,dt$

$= \int t + 2\,dt$

$= \frac{t^2}{2} + 2t + k$ → $v(t) = \frac{t^2}{2} + 2t + k$

$2{,}5 = \frac{1^2}{2} + 2.1 + k$

$2{,}5 = 2{,}5 + k$

$k = 2.5 - 2{,}5 = 0$

$s(t) = \int v(t)\,dt = \int \frac{t^2}{2} + 2t + 0$

$s(t) = \frac{1}{3} + \frac{t^3}{2} + \frac{2t^2}{2} + k$

$= \frac{t^3}{6} + t^2 + k$

$6 = \frac{t^3}{6} + 6^2 + k$

$6 = \frac{1^3}{6} + 1.2 + k$ x6

$36 = 1^2 + 6.6^2 + k$

$k = 36 - 6 = 30$

$s(t) = \frac{t^3}{6} + t^2 + 36$

b. Posisi benda saat $t = 2s$

$s(2) = \frac{2^3}{6} + 1^2 + 30$

$= \frac{8}{6} + \frac{24}{6} + \frac{180}{6} = \frac{212}{6} = 36{,}3$

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2. $L\, di/dt = 20t$ ampere dan pada $t=0$, $I = 5$, dan $q = 2$ C

Tentukan $i(t)$ dan $q(t)$ untuk $L = 0{,}1$ Henny dan grafiknya.

Berapa i dan q pada $t = 2$ sec

Jawab :

$L \frac{di}{dt} = 20t$ A ; $t = 0$, $i = 5$, $q = 2$

* $L(t) = 20 \times 0 = 0$

$i = \int 2\, dt = \int 20t\, dt$

$= 20 \frac{t^{1+1}}{1+1} + C$

$= 10t^2 + c$

$i(t) = 20t^2 + c$ → $i(0) = 5$

$5 = 10 \cdot 0^2 + c$

$c = 5$

$i(t) = 10t^2 + 5$

* $q = \int i\, dt = \int 10t^2 + 5\, dt$

$= 10 \frac{t^{2+1}}{2+1} + 5\frac{t^{0+1}}{0+1} + C$

$= \frac{10}{3} t^3 + 5t + c$

$q(t) = \frac{10}{3} t^3 + 5t + C$ → $q(0) = 2$

$2 = \frac{10}{3} \cdot 0^3 + 5 \cdot 0 + C$

$C = 2$

$q(t) = \frac{10}{3} t^3 + 5t + 2$

* $L = 0{,}1$ henny

$L(t) = 20t$

$0{,}1 = 20t$

$t = 0{,}005$

$i = \int 2\, dt = \int 20t\, dt$

$= 20 \frac{t^{1+1}}{1+1} + C$

$= 10t^2 + c$

$i(0{,}005) = 10(0{,}005)^2 + c$

$5 = 0{,}00025 + C$

$C = 4{,}99975$

$i(t) = 10t^2 + 4{,}99975$

* $q = \int i\, dt = \int 10t^2 + 4{,}99975\, dt$

$= 10 \frac{t^{2+1}}{2+1} + 4{,}99975 \frac{t^{0+1}}{0+1} + C$

~~$q(t) = \frac{10}{3} t^3 + 4{,}99975t + c$~~

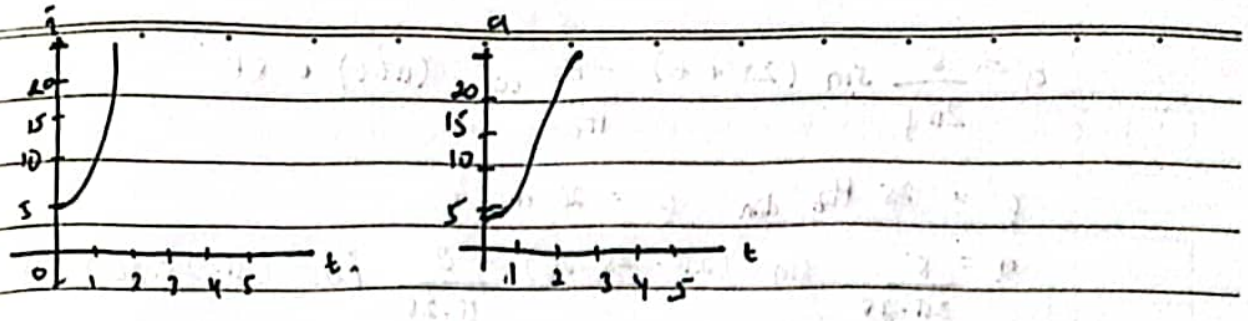
$q(0{,}005) = 10/3\,(0{,}005^3 + 4{,}99975 \cdot 0{,}0005 + c \quad (c \simeq 1{,}975)$

$q(t) = \frac{10}{3} t^3 + 4{,}99975 + 1{,}975$ . . . . . .

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3. Arus sebuah rangkaian listrik (AC) untuk setiap saat t memenuhi persamaan $i = 5\cos 2\pi t + 8\sin \pi t + 5$. Jika $f = 25$ Hz, hitung arus dan muatan pada $t = 20$ ms

* $I = 5\cos(2\pi \cdot 25 \cdot 20) + 8\sin(\pi \cdot 25 \cdot 20) + 5$

$= 5\cos(1000\pi) + 8\sin(500\pi) + 5$

$= 5 \cdot 2 + 8 \cdot 0 + 5$

$I = 10$ Ampere

* $q = \int i\,dt = \int 5\cos(2\pi + t) + 8\sin(\pi + t) + 5\,dt$

$= \int 5\cos(2\pi + t)\,dt + \int 8\sin(\pi f t)\,dt + \int 5\,dt$

(i) $\int 5\cos(2\pi + t)\,dt + \int 8\sin(\pi f t)\,dt + 5\,dt$

(i) $= \int 5\cos(2\pi + t)\,dt$

misal $a = 2\pi + t \rightarrow a' = 2\pi t$

$dt = \frac{da}{a'}$

$\int 5\cos a \frac{da}{2\pi f} = \frac{5}{2\pi f}\int \cos a\,da$

$= \frac{5}{2\pi f}\sin a = \frac{5}{2\pi f}\sin(2\pi + t)$

(ii) $\int 8\sin(\pi f t)\,dt$, misal $b = \pi f t \rightarrow b' = \pi t$

$db = \frac{db}{b'}$

$\int 8\sin b \frac{db}{\pi f} = \frac{8}{\pi f}\int \sin b\,db = \frac{-8}{\pi f}\cos b$

$= \frac{-8}{\pi f}\cos(\pi f t)$

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$q = \frac{5}{2\pi f} \sin(2\pi + t) - \frac{8}{\pi t} \cos(\pi t) + 5t$

$f = 25$ Hz dan $f = 20$ ms

$q = \frac{5}{2\pi \cdot 25} \sin(2\pi \cdot 25 \cdot 20) - \frac{8}{\pi \cdot 25} \cos(\pi \cdot 25 \cdot 20) + 5 \cdot 20$

$= \frac{5}{50\pi} \sin(1000\pi) - \frac{8}{25\pi} \cos(500\pi) + 100$

$= \frac{5}{50\pi} \cdot 0 - \frac{8}{25\pi} \cdot 1 + 100$

$= \frac{5}{50\pi} \cdot 0 - \frac{8}{25\pi} \cdot 1 + 100$

$q \simeq 99,8984$ Coulomb

9. Ans $i = 100(1 - e^{-0,2t})$ Tentukan i dan q steady state dan grafik

- $i = 100(1 - e^{-0,2t})$

Saat $t = 0 \rightarrow i = 100(1 - e^{-0,2(0)})$

$= 100(0)$

$= 0$ A

Saat $t = 1000 \rightarrow i = 100(1 - e^{-52(1000)})$

$= 100(1 - e^{-200})$

$= 100(1 - 1,38 \cdot 10^{-87}) \simeq 0$

$= 100(1)$

$= 100$ A

- $Q(t) = \int i(t)\, dt$

$= \int (100 - 100 e^{-0,2t})\, dt$

$= \int 100\, dt - \int 100 e^{-0,2t}\, dt$

$= \int 100\, dt - \int 100 \frac{1}{e^{0,2t}}\, dt$

$= 100t - \frac{100}{\frac{1}{0,2} e^{0,2t}} + C$

$\rightarrow 100t - 20 e^{-0,2t} + C$

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